# A complete fifteen-variable consistency equation

Strictly positive integer unknowns; exact degree 157; a 66-operation evaluator.

The only unknowns are

$$J, F, \alpha, z, f, h, i, T, s, w, \tau, \eta, \zeta, y, \gamma \in \mathbb{Z}_{>0}.$$

All other symbols in the equation are polynomial abbreviations:

$$\begin{gathered}
q_- = \beta J, \quad q = q_- + 1, \quad C = q - F - \alpha, \\
X = wq, \quad Y = sq^3, \quad E = XY, \quad k = \eta + \zeta, \\
\ell = EkY, \quad c = kY + \eta, \quad a = E + Y, \\
H = 4a + 3, \quad \Delta = a^2 + H, \quad D = X + ac + \gamma H, \\
R = [q_-(q - F) + \alpha](q^2 - 1) + (M_C + qM_F)J, \\
V = f^2(cT - R) - c, \quad S = i\Delta c^2, \\
N_0 = \tau^2 - \ell(\ell + k), \quad N_1 = D^2 - \Delta c^2, \\
N_a = S^2(V^2 - y^2) + y^2, \quad N_k = k - hE - R, \\
N_t = (K + w)C + (q - F) - zq_-, \quad N_s = \Delta f^2 - S^2.
\end{gathered}$$

The single equation is

$$\boxed{N_0 N_1 N_a N_k N_t N_s - \Delta = 0.}$$

**The four fixed integers.** Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\begin{gathered}
\rho = 2^{67}, \quad B = \rho^{8807}, \quad \beta = B - 1, \\
G_m(x) = \sum\nolimits_{j=0}^{m-1} x^j, \quad \Lambda = \rho^{4393} + \rho^{5495}.
\end{gathered}$$

$$\begin{aligned}
M_C &= \beta + \rho - (\rho + 3)\rho^{1063} \\
&\quad - \sum\nolimits_{e=0}^{3296} \lfloor (D_\rho(\Phi(\rho), e) + \rho - 1)/\rho \rfloor \, \rho^e, \\
M_F &= B - 1 + 4 + (2^{51} - 2)\Lambda \\
&\quad + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\
&\quad + (2^{65} - 1)\rho^{8792} G_{13}(\rho) + (2^{19} - 1)\rho^{8805}, \\
K &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\
&\quad + \Lambda\Phi(1/\rho),
\end{aligned}$$

where

$$\begin{aligned}
\Phi(x) &= (1 + 2^{51}) G_{943}(x) - 2^{10} \sum\nolimits_{i=0}^{942} v(t_i) x^i \\
&\quad + 2^{10} x^{1003} (x^{20} + 2^{17}) G_8(2x) \\
&\quad + 2^{18} x^{983} (x^{30} + 2^{17}) G_9(2x) \\
&\quad + 2^{44} x^{992} (x^{39} + 2x^{30} + 4x^{19} + 8) \\
&\quad + 2^{48} x^{1095} (1 + 2x^{1100} + 4x^{2201}).
\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

**The 943 indexed fixed tiles.** Use the actions $u_j, v_j, d_j$ decoded below. Let

$$\begin{gathered}
h(q, b) = 2(q + 1) + b, \quad \mathcal{R} = \{u_j : d_j = 1, \ 0 \le j < 430\}, \\
\mathcal{L} = \{u_j : d_j = 0, \ 0 \le j < 430\}, \\
r_q = 6 + |\mathcal{R} \cap \{0, \dots, q - 1\}|, \quad l_q = 205 + |\mathcal{L} \cap \{0, \dots, q - 1\}|.
\end{gathered}$$

An interval ending at $-1$ is empty. The table defines $t_i$ for $0 \le i \le 942$; all index ranges are fixed:

| index $i$ | tile $t_i = (w, n, e, s)$ |
|---|---|
| $0, 1$ | $(2, 435, 3, 1), \ (5, 0, 0, 0)$ |
| $j + 2 \ (j = 2q + b < 430)$ | $((1 - d_j) l_{u_j}, h(q, b), d_j r_{u_j}, v_j)$ |
| $432 + b \ (b = 0, 1)$ | $(0, h(215, b), 0, h(215, b))$ |
| $434 + b \ (b = 0, 1)$ | $(0, b, 0, b)$ |
| $424 + 2r_q + b \ (q \in \mathcal{R}, b = 0, 1)$ | $(r_q, b, 0, h(q, b))$ |
| $424 + 2l_q + b \ (q \in \mathcal{L}, b = 0, 1)$ | $(0, b, l_q, h(q, b))$ |
| $924 + j \ (0 \le j < 4)$ | $(0, j + 430 \lfloor j/2 \rfloor, 0, 438)$ |
| 928 | $(2, 434, 2, 0)$ |
| 929 | $(3, 436, 4, 4)$ |
| 930 | $(4, 436, 4, 0)$ |
| $497 + t \ (t = 434, 435, 436)$ | $(1, 438, 1, t)$ |
| $^{934+3a+b-6\lfloor a/4 \rfloor}_{a \in \{0,1,4\}, \ b=0,1,2}$ | $(a, 437, b, 437)$ |

For $t = (w, n, e, s)$ put

$$v(t) = w + 2^8 n + 2^{17} e + 2^{25} s + 127\, 2^{34}.$$

**A closed decoder for the 430 actions.** Put $D_b(A, k) = \lfloor A/b^k \rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$\begin{gathered}
N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \lfloor \mathcal{A}/2^{636} \rfloor \bmod 2^{1675}, \\
E_0 = \lfloor \mathcal{A}/2^{2311} \rfloor, \\
\nu_j = D_2(N, j), \quad \sigma_j = \sum\nolimits_{k=0}^{j-1} \nu_k, \\
m_q = D_2(N, 430 + q), \quad \tau_q = \sum\nolimits_{k=0}^{q-1} m_k, \\
F_q = 14(1 - m_q) + m_q D_{16}(E_0, \tau_q), \\
u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j) D_{216}(T_0, j - \sigma_j), \\
v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).
\end{gathered}$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

**The integer $\mathcal{A}$.** The alphabet, in order of digit values $0, \dots, 61$, is `0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz`. Concatenate the following 436 characters as digits $e_0, \dots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```

The ordinary state/color-conjugacy, native and Pell arguments give unsolvability iff Con(ZFC). Native calibration and internal-ZFC formalization remain open in Lean. This page is a complete finite specification, not an unsolvability proof.